CASEY MAIKALANI CRANE

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EDUCATION

BS, Mathematics, University of Texas at Dallas, 2015
MS, Mathematics, University of Texas at Dallas, 2020
PhD, Mathematics, University of Texas at Dallas, 2026

PUBLICATIONS

- "Consensus-breaking global Hopf bifurcation in memory-based multi-agent systems." PhD Dissertation. Advisor: Dr. Wieslaw Krawcewicz.

Journal Publications

- Ziad Ghanem, Casey Crane, Jingzhou Liu. "Global bifurcation of non-radial solutions for symmetric sub-linear elliptic systems on the planar unit disc." *Journal of Fixed Point Theory and Applications* 26.4 (2024): 44.
- Yameng Duan, Casey Crane, Wieslaw Krawcewicz, and Huafeng Xiao "Periodic solutions in reversible symmetric second order systems with multiple distributed delays" *Journal of Differential Equations* 401 (2024): 282–307.
- Chaoquan Chen, Casey Crane, and Travis Hensley. "Global Hopf bifurcation in symmetric configurations of distributed delay differential equations" *Communications in Pure and Applied Analysis* 401 (2025): 282–307.

Preprints

- Casey Crane. "Global Hopf bifurcation and symmetric periodic solutions in multi-agent aystems with pseudoneutral distributed delays" arXiv preprint arXiv:1711.02160 (2026). (submitted to SIADS)
- Casey Crane "Trust as hysteresis: analytically tractable strategies for the iterated prisoner's dilemma" arXiv preprint arXiv:2504.06519 (2026) (submitted to J. Theor. Biol.)
- Casey Crane and Ziad Ghanem. "Unbounded branches of non-radial solutions to semilinear elliptic systems on a disc and their patterns." arXiv preprint arXiv:2504.06519 (2025). (submitted to TMNA)

RESEARCH SUMMARY

My research uses equivariant topological methods to study symmetry-breaking and pattern formation across a wide variety of problems, with a focus on highly symmetric networked systems and delay equations. I strive to combine technical abstract results applicable to broad families of problems with specific concrete applications, supported by numerical analysis. I am particularly interested in using these techniques to explore self-organizing behaviors, the breakdown of consensus, as well as cooperation and competition in networks of intelligent autonomous agents.

In (Duan 2024), we show pattern formation for a general class of DDEs with arbirarily many distributed delays, and apply this to systems of geometrically coupled oscillators. In (Chen 2025), we showed spontaneous pattern formation in systems of distributed delay differential equations. We then explored particular applications in anti-windup PID controllers and controllable pattern formation in geometric drone swarms. In "Multi-Agent Systems with Pseudoneutral Distributed Delays" (preprint), we showed how consensus breakdown in a multi-agent system where agents have memory of their states and decisions leads to global bifurcation of periodic multiconsensus patterns, and showed an application modeling coupled asset markets with two populations of traders, also supported by numerical simulations.

My experience has taught me that the most exciting research lies in the intersection of different fields, and I am highly adaptable. I am equally comfortable driving my own research agenda and collaborating deeply to advance the goals of a larger research group, and I am always looking for opportunities to broaden my horizons.

EXPERIENCE

Teaching Assistant, University of Texas at Dallas — *Fall 2021 – Fall 2025*
Teaching Associate, University of Texas at Dallas — *Spring 2026*
Systems and Processes Engineer, Good Faith Energy, Addison, Texas — *2020 – 2024*

CONFERENCE PRESENTATIONS

- SIAM Texas-Louisiana Section Annual Meeting, Baylor University, 2024
- Talk "Global symmetric Hopf bifurcation using equivariant degree," Hunan Normal University, 2023

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